

Khoa Điện tử - Viễn thông**Bộ môn Điện tử****Môn Học Thiết kế hệ thống trên chip và kiểm định thiết kế****Báo cáo Thực Hành****Lab 02: Design Planning (Xây dựng Floorplan và đường Power)****Họ tên SV1: Bùi Hồng Hà****Họ tên SV2: Trần Vũ Bảo****MSSV1: 22200050****MSSV2: 22200018****Sinh viên trả lời các đề mục sau khi thực hiện bài Lab tương ứng:****1. Data setup:****-Công nghệ (tên và process):**

```
icc2_shell> get_libs  
{ORCA_TOP.dlib saed32_lp9m_tech saed32_hvt saed32_lvt saed32_rvt saed32_sram_lp}
```

-Số lớp metal:

```
icc2_shell> get_layers M*  
{M1 M2 M3 M4 M5 M6 M7 M8 M9 MRDL M1PIN M2PIN M3PIN M4PIN M5PIN M6PIN M7PIN M8PIN M9PIN MRDL9PIN}
```

Tổng cộng 20 lớp metal.**-Tần số hoạt động chip/block:**

- Block ORCA_TOP có tổng cộng 11 master clocks. Tần số tương ứng ($f = 1/\text{Period}$) của các clock chính như sau:
 - PCI_CLK: $T = 7.50 \text{ ns} \rightarrow f \approx 133.33 \text{ MHz}$
 - SDRAM_CLK: $T = 4.10 \text{ ns} \rightarrow f \approx 243.90 \text{ MHz}$
 - SD_DDR_CLK / SD_DDR_CLKn: $T = 4.10 \text{ ns} \rightarrow f \approx 243.90 \text{ MHz}$
 - SYS_2x_CLK: $T = 2.30 \text{ ns} \rightarrow f \approx 434.78 \text{ MHz}$ (clock nhanh nhất)

- SYS_CLK: $T = 4.60\text{ ns} \rightarrow f \approx 217.39\text{ MHz}$ (sinh ra từ SYS_2x_CLK qua bộ chia 2)

```
icc2_shell> report_clock
*****
Report : clock
Design : ORCA_TOP
Mode : func
Version: 0-2018.06-SP1
Date : Tue May 5 13:38:32 2026
*****

Attributes:
  p - Propagated clock
  G - Generated clock
  U - Unexpanded generated clock

Clock      Period  Waveform      Attrs  Sources
-----
PCI_CLK      7.50   {0 3.75}
SDRAM_CLK    4.10   {0 2.05}
SD_DDR_CLK   4.10   {0 2.05}      G      {sd_CLK}
SD_DDR_CLKn  4.10   {2.05 4.1}    G      {sd_CLKn}
SYS_2x_CLK   2.30   {0 1.15}
SYS_CLK      4.60   {0 2.3}      G      {I_CLOCKING/sys_clk_in_reg/Q}
v_PCI_CLK    7.50   {0 3.75}
v_SDRAM_CLK  4.10   {0 2.05}
{}

Generated  Master      Generated      Master      Waveform
Clock      Source      Source      Clock      Modification
-----
SD_DDR_CLK  sdram_clk   sd_CLK      SDRAM_CLK  div(1), combinational
SD_DDR_CLKn sdram_clk   sd_CLKn     SDRAM_CLK  div(1), combinational, inv
SYS_CLK     sys_2x_clk  I_CLOCKING/sys_clk_in_reg/Q
              SYS_2x_CLK  div(2)
```

-Các mức điện áp và nhiệt độ hoạt động (*report_corners*):

```
icc2_shell> report_corners
*****
Report : corner
Design : ORCA_TOP
Version: 0-2018.06-SP1
Date   : Tue May  5 13:39:41 2026
*****

-----
Corner ff_125c
-----

Current: false Default: false Empty: false

Scenarios associated with this corner:

Scenario      Mode      Active Setup Hold Leakage Dynamic Max_tran Max_cap Min_cap Cell_em Signal_em
-----
func.ff_125c   func      true  false true  false  false   true   true   true   false  false
test.ff_125c   test      true  false true  false  false   true   true   true   false  false

Top-Level PVT Settings:
early process label      fast
early process number     1.01
early voltage            0.95
source                   UPF
file/line                 /global/gtsna_training/projects/IC_Compiler_II_Implementation/dev_2018.06/ref/ORCA_TOP_constraints/ORCA_TOP_c_ff_125c.tcl, line 6
early temperature        125.00
source                   hierarchical setting
file/line                 /global/gtsna_training/projects/IC_Compiler_II_Implementation/dev_2018.06/ref/ORCA_TOP_constraints/ORCA_TOP_c_ff_125c.tcl, line 3

late process label      fast
late process number     1.01
late voltage            0.95
source                   UPF
file/line                 /global/gtsna_training/projects/IC_Compiler_II_Implementation/dev_2018.06/ref/ORCA_TOP_constraints/ORCA_TOP_c_ff_125c.tcl, line 6
late temperature        125.00
source                   hierarchical setting
file/line                 /global/gtsna_training/projects/IC_Compiler_II_Implementation/dev_2018.06/ref/ORCA_TOP_constraints/ORCA_TOP_c_ff_125c.tcl, line 3

0 objects with explicit PVT settings.

Top-level cell timing derates:
data_cell derate early:  rise: 1.00  fall: 1.00
clk_cell derate early:   rise: 1.00  fall: 1.00
cell_check derate early: rise: 1.00  fall: 1.00
data_cell derate late:   rise: 1.05  fall: 1.05
clk_cell derate late:    rise: 1.05  fall: 1.05
cell_check derate late:  rise: 1.00  fall: 1.00

Global net timing derates:
net_data, early:  rise: 1.00  fall: 1.00
net_data, late:   rise: 1.05  fall: 1.05
net_clock, early: rise: 1.00  fall: 1.00
net_clock, late:  rise: 1.05  fall: 1.05

-----
Corner ff_m40c
-----

Current: false Default: false Empty: false

Scenarios associated with this corner:

Scenario      Mode      Active Setup Hold Leakage Dynamic Max_tran Max_cap Min_cap Cell_em Signal_em
-----
func.ff_m40c   func      true  false true  false  false   true   true   true   false  false

Top-Level PVT Settings:
early process label      fast
early process number     1.01
early voltage            0.95
source                   UPF
file/line                 /global/gtsna_training/projects/IC_Compiler_II_Implementation/dev_2018.06/ref/ORCA_TOP_constraints/ORCA_TOP_c_ff_m40c.tcl, line 6
early temperature        -40.00
source                   hierarchical setting
file/line                 /global/gtsna_training/projects/IC_Compiler_II_Implementation/dev_2018.06/ref/ORCA_TOP_constraints/ORCA_TOP_c_ff_m40c.tcl, line 3

late process label      fast
late process number     1.01
late voltage            0.95
source                   UPF
file/line                 /global/gtsna_training/projects/IC_Compiler_II_Implementation/dev_2018.06/ref/ORCA_TOP_constraints/ORCA_TOP_c_ff_m40c.tcl, line 6
late temperature        -40.00
source                   hierarchical setting
file/line                 /global/gtsna_training/projects/IC_Compiler_II_Implementation/dev_2018.06/ref/ORCA_TOP_constraints/ORCA_TOP_c_ff_m40c.tcl, line 3
```

```
0 objects with explicit PVT settings.

Top-level cell timing derates:
data_cell derate early:  rise: 1.00  fall: 1.00
clk_cell derate early:   rise: 1.00  fall: 1.00
cell_check derate early: rise: 1.00  fall: 1.00
data_cell derate late:   rise: 1.05  fall: 1.05
clk_cell derate late:    rise: 1.05  fall: 1.05
cell_check derate late:  rise: 1.00  fall: 1.00

Global net timing derates:
net_data, early:  rise: 1.00  fall: 1.00
net_data, late:   rise: 1.05  fall: 1.05
net_clock, early: rise: 1.00  fall: 1.00
net_clock, late:  rise: 1.05  fall: 1.05

-----
Corner ss_125c
-----

Current: true Default: false Empty: false

Scenarios associated with this corner:

Scenario      Mode      Active Setup Hold Leakage Dynamic Max_tran Max_cap Min_cap Cell_em Signal_em
-----
func.ss_125c  func      true  true  false true      true      true      true      true      false  false
test.ss_125c  test      true  true  false false     false     true      true      true      false  false

Top-Level PVT Settings:
early process label      slow
early process number     0.99
early voltage            0.75
source                   UPF
file/line                 /global/gtsna_training/projects/IC_Compiler_II_Implementation/dev_2018.06/ref/ORCA_TOP_constraints/ORCA_TOP_c_ss_125c.tcl, line 6
early temperature       125.00
source                   hierarchical setting
file/line                 /global/gtsna_training/projects/IC_Compiler_II_Implementation/dev_2018.06/ref/ORCA_TOP_constraints/ORCA_TOP_c_ss_125c.tcl, line 3

late process label      slow
late process number     0.99
late voltage            0.75
source                   UPF
file/line                 /global/gtsna_training/projects/IC_Compiler_II_Implementation/dev_2018.06/ref/ORCA_TOP_constraints/ORCA_TOP_c_ss_125c.tcl, line 6
late temperature       125.00
source                   hierarchical setting
file/line                 /global/gtsna_training/projects/IC_Compiler_II_Implementation/dev_2018.06/ref/ORCA_TOP_constraints/ORCA_TOP_c_ss_125c.tcl, line 3

0 objects with explicit PVT settings.

Top-level cell timing derates:
data_cell derate early:  rise: 0.93  fall: 0.93
clk_cell derate early:   rise: 0.93  fall: 0.93
cell_check derate early: rise: 1.00  fall: 1.00
data_cell derate late:   rise: 1.00  fall: 1.00
clk_cell derate late:    rise: 1.00  fall: 1.00
cell_check derate late:  rise: 1.00  fall: 1.00

Global net timing derates:
net_data, early:  rise: 0.93  fall: 0.93
net_data, late:   rise: 1.00  fall: 1.00
net_clock, early: rise: 0.93  fall: 0.93
net_clock, late:  rise: 1.00  fall: 1.00
```

2. Xây dựng Floorplan:

-Cho biết boundary của block (*get_attribute -name boundary -objects [current_block]*)

icc2_shell> get_attribute -name boundary -objects [current_block]
{0.0000 0.0000} {0.0000 539.9280} {501.4480 539.9280} {501.4480 789.0560} {1042.8960 789.0560} {1042.8960 0.0000}

-Cho biết diện tích của block (Core Area và Chip Area) (*report_design*)

Core Area : 626300.529
Chip Area : 697978.609

-Cho biết Utilization (*report_utilization*):

```
icc2_shell> report_utilization
*****
Report : report_utilization
Design : ORCA_TOP
Version: 0-2018.06-SP1
Date   : Tue May  5 13:45:42 2026
*****
Utilization Ratio:                                0.4916
Utilization options:
- Area calculation based on:                      site_row of block ORCA_TOP/floorplan
- Categories of objects excluded:                  hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                                         626300.5288
Total Capacity Area:                              356544.9712
Total Area of cells:                              175285.9124
Area of excluded objects:
- hard_macros      :                               269755.5576
- macro_keepouts   :                               0.0000
- soft_macros      :                               0.0000
- io_cells         :                               0.0000
- hard_blockages   :                               0.0000

Utilization of site-rows with:
- Site 'unit':                                       0.4916

0.4916
```

Giải thích Utilization là gì?

- Utilization (mật độ sử dụng) là tỷ lệ giữa tổng diện tích các standard cells đã được đặt (placed) so với tổng diện tích core khả dụng (sau khi trừ các hard macros, macro keepouts, soft macros, IO cells và hard blockages).
- Công thức:

$$\text{Utilization} = (\text{Tổng diện tích standard cells}) / (\text{Diện tích core còn lại sau khi trừ macros và blockages})$$
- Với thiết kế ORCA_TOP, utilization = 0.4916 ($\approx 49.16\%$), nằm trong khoảng hợp lý ($\sim 50\%$) cho giai đoạn floorplan, còn đủ khoảng trống cho placement, CTS và routing ở các bước tiếp theo.

-Số macros và tên các macros:

- Tổng số macros là 40

```
icc2_shell> sizeof_collection [get_flat_cells -filter "is_hard_macro==true"]
40
```

```
icc2_shell> get_attribute [get_flat_cells -filter "is_hard_macro==true"] full_name
```

```
I_PCI_TOP/I_PCI_READ_FIFO/PCI_FIFO_RAM_8 I_PCI_TOP/I_PCI_READ_FIFO/PCI_FIFO_RAM_7 I_PCI_TOP/I_PCI_READ_FIFO/PCI_FIFO_RAM_6 I_PCI_TOP/I_PCI_READ_FIFO/PCI_FIFO_RAM_5 I_PCI_TOP/I_PCI_READ_FIFO/PCI_FIFO_RAM_4
I_PCI_TOP/I_PCI_READ_FIFO/PCI_FIFO_RAM_3 I_PCI_TOP/I_PCI_READ_FIFO/PCI_FIFO_RAM_2 I_PCI_TOP/I_PCI_READ_FIFO/PCI_FIFO_RAM_1 I_PCI_TOP/I_PCI_WRITE_FIFO/PCI_FIFO_RAM_8 I_PCI_TOP/I_PCI_WRITE_FIFO/PCI_FIFO_RAM_7
I_PCI_TOP/I_PCI_WRITE_FIFO/PCI_FIFO_RAM_6 I_PCI_TOP/I_PCI_WRITE_FIFO/PCI_FIFO_RAM_5 I_PCI_TOP/I_PCI_WRITE_FIFO/PCI_FIFO_RAM_4 I_PCI_TOP/I_PCI_WRITE_FIFO/PCI_FIFO_RAM_3 I_PCI_TOP/I_PCI_WRITE_FIFO/PCI_FIFO_RAM_2
I_PCI_TOP/I_PCI_WRITE_FIFO/PCI_FIFO_RAM_1 I_CONTEXT_MEM/I_CONTEXT_RAM_0_1 I_CONTEXT_MEM/I_CONTEXT_RAM_0_2 I_CONTEXT_MEM/I_CONTEXT_RAM_0_3 I_CONTEXT_MEM/I_CONTEXT_RAM_0_4 I_CONTEXT_MEM/I_CONTEXT_RAM_1_1 I_CONTEXT_MEM/I_CONTEXT_RAM_1_2
I_CONTEXT_MEM/I_CONTEXT_RAM_1_3 I_CONTEXT_MEM/I_CONTEXT_RAM_1_4 I_CONTEXT_MEM/I_CONTEXT_RAM_2_1 I_CONTEXT_MEM/I_CONTEXT_RAM_2_2 I_CONTEXT_MEM/I_CONTEXT_RAM_2_3 I_CONTEXT_MEM/I_CONTEXT_RAM_2_4 I_CONTEXT_MEM/I_CONTEXT_RAM_3_1
I_CONTEXT_MEM/I_CONTEXT_RAM_3_2 I_CONTEXT_MEM/I_CONTEXT_RAM_3_3 I_CONTEXT_MEM/I_CONTEXT_RAM_3_4 I_RISC_CORE/I_REG_FILE/REG_FILE_D_RAM I_RISC_CORE/I_REG_FILE/REG_FILE_C_RAM I_RISC_CORE/I_REG_FILE/REG_FILE_B_RAM
I_RISC_CORE/I_REG_FILE/REG_FILE_A_RAM I_SDRAM_TOP/I_SDRAM_READ_FIFO/SD_FIFO_RAM_1 I_SDRAM_TOP/I_SDRAM_READ_FIFO/SD_FIFO_RAM_0 I_SDRAM_TOP/I_SDRAM_WRITE_FIFO/SD_FIFO_RAM_1 I_SDRAM_TOP/I_SDRAM_WRITE_FIFO/SD_FIFO_RAM_0
```

-Số block IOs:

- Tổng số ports (IOs) của block ORCA_TOP là 237 ports, bao gồm cả ports dữ liệu tín hiệu (data, address, control) và ports clock, reset, scan, test mode...

```
icc2_shell> sizeof_collection [get_port]
237
```

```
icc2_shell> get_port
```

```
{sdram_clk sys_2x_clk shutdown test_mode {test_si[5]} {test_si[4]} {test_si[3]} {test_si[2]} {test_si[1]} {test_si[0]} {test_so[5]} {test_so[4]} {test_so[3]} {test_so[2]} {test_so[1]} {test_so[0]} scan_enable ate_clk occ_bypass
occ_reset pclk prst_n pidlel pght_n {pad_in[31]} {pad_in[30]} {pad_in[29]} {pad_in[28]} {pad_in[27]} {pad_in[26]} {pad_in[25]} {pad_in[24]} {pad_in[23]} {pad_in[22]} {pad_in[21]} {pad_in[20]} {pad_in[19]} {pad_in[18]} {pad_in[17]}
{pad_in[16]} {pad_in[15]} {pad_in[14]} {pad_in[13]} {pad_in[12]} {pad_in[11]} {pad_in[10]} {pad_in[9]} {pad_in[8]} {pad_in[7]} {pad_in[6]} {pad_in[5]} {pad_in[4]} {pad_in[3]} {pad_in[2]} {pad_in[1]} {pad_in[0]} {pad_out[31]}
{pad_out[30]} {pad_out[29]} {pad_out[28]} {pad_out[27]} {pad_out[26]} {pad_out[25]} {pad_out[24]} {pad_out[23]} {pad_out[22]} {pad_out[21]} {pad_out[20]} {pad_out[19]} {pad_out[18]} {pad_out[17]} {pad_out[16]} {pad_out[15]}
{pad_out[14]} {pad_out[13]} {pad_out[12]} {pad_out[11]} {pad_out[10]} {pad_out[9]} {pad_out[8]} {pad_out[7]} {pad_out[6]} {pad_out[5]} {pad_out[4]} {pad_out[3]} {pad_out[2]} {pad_out[1]} {pad_out[0]} pad_en ppar_in ppar_out ppar_en
{pc_be_in[3]} {pc_be_in[2]} {pc_be_in[1]} {pc_be_in[0]} {pc_be_out[3]} {pc_be_out[2]} {pc_be_out[1]} {pc_be_out[0]} ...}
```

-Cho biết Congestion là gì?

- Congestion (mật độ routing) là tình trạng số lượng wires cần đi qua một vùng (GRC – global routing cell) vượt quá số tracks routing khả dụng của vùng đó. Hiệu số giữa yêu cầu và cung cấp được gọi là overflow: $\text{overflow} > 0$ nghĩa là tắc nghẽn, có nguy cơ routing không thành công hoặc phải đi vòng. ICC2 đánh giá congestion theo ba tiêu chí: tổng cả hai hướng (Both Dirs), riêng hướng ngang (H routing) và riêng hướng dọc (V routing).

Cho biết tỉ lệ overflow của Both Dirs/H Routing/V Routing trong design là bao nhiêu? (*report_congestion*).

```
icc2_shell> report_congestion
```

```
*****
```

```
Report : congestion
```

```
Design : ORCA_TOP
```

```
Version: 0-2018.06-SP1
```

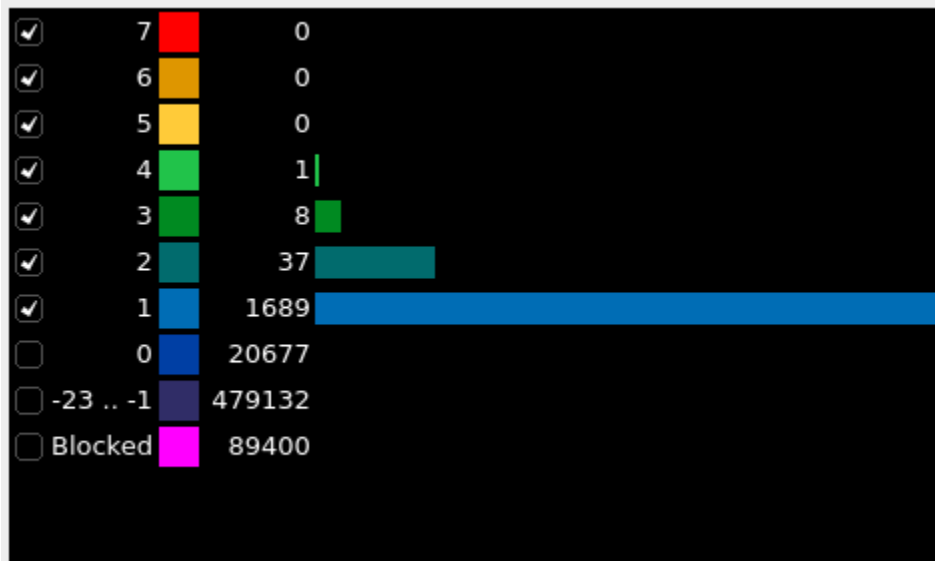
```
Date : Tue May 5 14:01:31 2026
```

```
*****
```

Layer	overflow		# GRCs has	
Name	total	max	overflow (%)	max overflow

Both Dirs	1791	4	1735 (0.29%)	1
H routing	1741	4	1701 (0.58%)	1
V routing	50	3	34 (0.01%)	4

Hình ảnh Congestion Map.



-Hình ảnh floorplan chip (chỉ hiển thị shape, IOs và các macros/block)



3. Xây dựng đường Power:

-Số power domain và tên:

```
icc2_shell> get_power_domains
{PD_ORCA_TOP PD_RISC_CORE}
```

- Do đó có 2 power domain là PD_ORCA_TOP và PD_RISC_CORE.

```
icc2_shell> report_power_domains
*****
Report : Power Domain
Design : ORCA_TOP
Corner : ss_125c
Version: 0-2018.06-SP1
Date   : Tue May  5 14:12:46 2026
*****
-----
Power Domain      : PD_ORCA_TOP
Current Scope     : / (top scope)
Elements         : ORCA_TOP
Voltage Area      : DEFAULT_VA
Available Supply Nets : VDD, VDDH, VSS
Available Supply Sets  : ss_high, ss_low

Default Supplies      - Power -      - Ground -
Primary              : VDD [0.75]    VSS [0.00]
Isolation            : --
Retention            : --

Power Switch         :
Isolation Strategy   :
Level Shifter Strategy :
Repeater Strategy    :
Retention Strategy   :
Always on strategy   : dual_power
-----
Power Domain      : PD_RISC_CORE
Current Scope     : / (top scope)
Elements         : I_RISC_CORE
Voltage Area      : PD_RISC_CORE
Available Supply Nets : VDD, VDDH, VSS
Available Supply Sets  : ss_high, ss_low

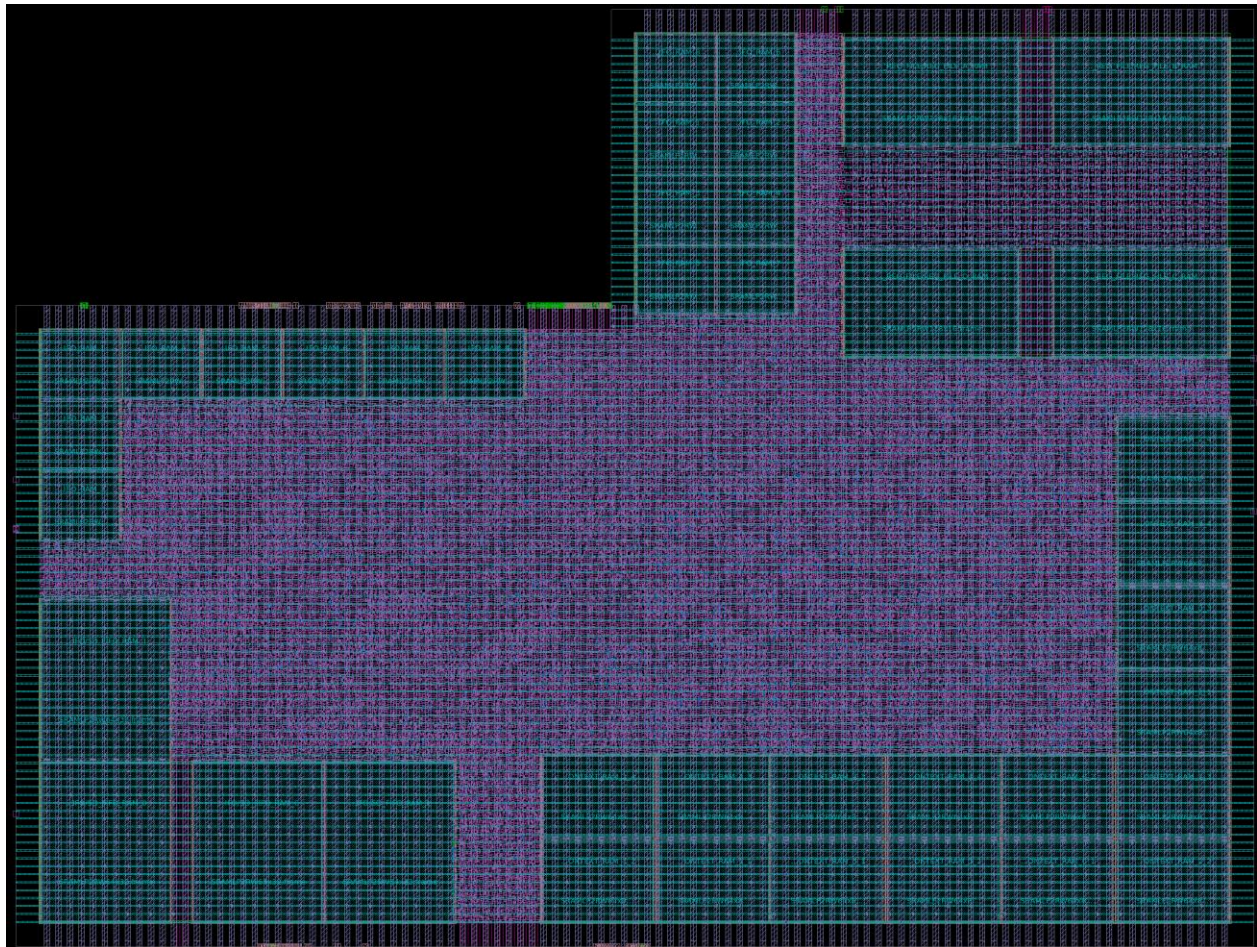
Default Supplies      - Power -      - Ground -
Primary              : VDDH [0.95]    VSS [0.00]
Isolation            : --
Retention            : --

Power Switch         :
Isolation Strategy   :
Level Shifter Strategy : risc_ls_in
Applies To          : inputs
Rule                : low_to_high
Location            : self
Elements            : N/A
Excluded Elements    : N/A
Mapped Cells        : N/A

Level Shifter Strategy : risc_ls_out
Applies To          : outputs
Rule                : high_to_low
Location            : parent
Elements            : N/A
Excluded Elements    : N/A
Mapped Cells        : N/A

Repeater Strategy    :
Retention Strategy   :
Always on strategy   : dual_power
```


-Hình ảnh chip sau khi xây dựng power?



-Cho biết các lệnh để tạo và kết nối power Ring, Mesh (straps) and Rail? Cho ví dụ.

- Lệnh tạo, kết nối power Mesh và ví dụ:

```
create_pg_mesh_pattern P_top_two \
-layers { \
  { {horizontal_layer: M7} {width: 1.104} {spacing: interleaving} {pitch: 13.376} {offset: 0.856} {trim : true} } \
  { {vertical_layer: M8} {width: 4.64} {spacing: interleaving} {pitch: 19.456} {offset: 6.08} {trim : true} } \
} \
-via_rule { {intersection: adjacent} {via_master : pgvia_8x10} }

# m2 pitch=0.152; 0.152*48=7.296
```

- Lệnh tạo, kết nối power Ring và ví dụ:

```
create_pg_ring_pattern P_HM_ring -horizontal_layer M5 -horizontal_width {1} -vertical_layer M6 -vertical_width {1} -corner_bridge false
set_pg_strategy S_HM_ring_top -macros $hm(top) -pattern { {pattern: P_HM_ring} {nets: {VSS VDD}} {offset: {0.3 0.3}} }
set_pg_strategy S_HM_ring_risc -macros $hm(risc_core) -pattern { {pattern: P_HM_ring} {nets: {VSS VDDH}} {offset: {0.3 0.3}} }
set_pg_strategy via_rule S_ring_vias -via_rule { \
  {{{strategies: {S_HM_ring_top S_HM_ring_risc}} {layers: {M5}}} {existing: {strap }}{via_master: {default}} } \
  {{{strategies: {S_HM_ring_top S_HM_ring_risc}} {layers: {M6}}} {existing: {strap }}{via_master: {default}} } \
}
```

- Lệnh tạo, kết nối power Rail và ví dụ:

```
create_pg_std_cell_conn_pattern P_std_cell_rail

set_pg_strategy S_std_cell_rail_VSS_VDD \
    -core \
    -blockage { {{nets: VDD} {voltage_areas: PD_RISC_CORE}} {macros_with_keepout: $all_macros} } \
    -pattern {{pattern: P_std_cell_rail}{nets: {VSS VDD}}} \
    -extension {{stop: outermost_ring}{direction: L B R T }}

set_pg_strategy S_std_cell_rail_VDDH \
    -voltage_areas PD_RISC_CORE \
    -blockage {macros_with_keepout: $all_macros} \
    -pattern {{pattern: P_std_cell_rail}{nets: {VDDH}}}

set_pg_strategy_via_rule S_via_stdcellrail \
    -via_rule {{intersection: adjacent}{via_master: default}}
```

-Cho biết các thông số xây dựng power straps (số đường straps, độ rộng, khoảng cách):

- Số đường straps:

```
icc2_shell> sizeof_collection [get_shapes -filter {net_type==power}]
1419
```

```
create_pg_mesh_pattern P_top_two \
    -layers { \
        { {horizontal_layer: M7} {width: 1.104} {spacing: interleaving} {pitch: 13.376} {offset: 0.856} {trim : true} } \
        { {vertical_layer: M8} {width: 4.64 } {spacing: interleaving} {pitch: 19.456} {offset: 6.08} {trim : true} } \
    } \
    -via_rule { {intersection: adjacent} {via_master : pgvia_8x10} }

# m2 pitch=0.152; 0.152*48=7.296
create_pg_mesh_pattern P_m2_triple \
    -layers { \
        { {vertical_layer: M2} {track_alignment : track} {width: 0.44 0.192 0.192} {spacing: 2.724 3.456} {pitch: 9.728} {offset: 1.216} {trim : true} } \
    } }
```

Thiết kế sử dụng 2 bộ mesh pattern song song:

+ Mesh phía trên (P_top_two) gồm 2 lớp straps:

- Lớp ngang M7: width = 1.104 μm , pitch = 13.376 μm , spacing = interleaving, offset = 0.856 μm .

- Lớp dọc M8: width = 4.64 μm , pitch = 19.456 μm , spacing = interleaving, offset = 6.08 μm .

+ Mesh trung gian (P_m2_triple) trên lớp M2: width = 0.44 / 0.192 / 0.192 μm (multi-width), pitch = 9.728 μm , spacing = 2.724 / 3.456 μm , offset = 1.216 μm .

+ Power ring (P_HM_ring) bao quanh top và risc_core: lớp ngang M5 và lớp dọc M6, width = 1 μm cả hai chiều, offset 0.3 μm .

+ Standard cell rail trên M1 (P_std_cell_rail) phủ toàn core, nối trực tiếp VDD/VSS/VDDH cho cả hai voltage area DEFAULT_VA và PD_RISC_CORE.

- Cho biết kết quả các lệnh kiểm tra sau khi xây dựng đường power (*check_pg_missing_vias*, *check_pg_drc*, *check_pg_connectivity*)

- Lệnh *check_pg_missing_vias*:

```
icc2_shell> check_pg_missing_vias
Check net VDD vias...
Number of missing vias on VIA5 layer: 3
Number of missing vias on VIA6 layer: 1
Total number of missing vias: 4
Checking net VDD vias took 0 seconds.
Check net VDDH vias...
Number of missing vias on VIA6 layer: 13
Total number of missing vias: 13
Checking net VDDH vias took 0 seconds.
Check net VSS vias...
Number of missing vias on VIA6 layer: 1
Total number of missing vias: 1
Checking net VSS vias took 0 seconds.
Overall runtime: 0 seconds.
```

- Lệnh *check_pg_drc*:

```
icc2_shell> check_pg_drc
Command check_pg_drc started at Tue May 5 14:24:41 2026
Command check_pg_drc finished at Tue May 5 14:24:46 2026
CPU usage for check_pg_drc: 15.85 seconds ( 0.00 hours)
Elapsed time for check_pg_drc: 5.47 seconds ( 0.00 hours)
Total number of errors found: 903926
    16021 shorts on C0
    580113 shorts on M1
    65066 insufficient spacings on C0
    242726 insufficient spacings on M1
-----
Description of the errors can be seen in gui error set "DRC_report_by_check_pg_drc"
-----
```


- Lệnh *check_pg_connectivity*:

```
icc2_shell> check_pg_connectivity
```

```
Loading cell instances...
```

```
Number of Standard Cells: 52007
```

```
Number of Macro Cells: 40
```

```
Number of IO Pad Cells: 0
```

```
Number of Blocks: 0
```

```
Loading P/G wires and vias...
```

```
Number of VDD Wires: 971
```

```
Number of VDD Vias: 24111
```

```
Number of VDD Terminals: 198
```

```
Number of VDDH Wires: 211
```

```
Number of VDDH Vias: 2890
```

```
Number of VDDH Terminals: 39
```

```
Number of VSS Wires: 2853
```

```
Number of VSS Vias: 52048
```

```
Number of VSS Terminals: 263
```

```
*****Verify net VDD connectivity*****
```

```
Number of floating wires: 26
```

```
Number of floating vias: 0
```

```
Number of floating std cells: 38044
```

```
Number of floating hard macros: 0
```

```
Number of floating I/O pads: 0
```

```
Number of floating terminals: 0
```

```
Number of floating hierarchical blocks: 0
```

```
*****
```

```
*****Verify net VDDH connectivity*****
```

```
Number of floating wires: 0
```

```
Number of floating vias: 0
```

```
Number of floating std cells: 1076
```

```
Number of floating hard macros: 0
```

```
Number of floating I/O pads: 0
```

```
Number of floating terminals: 0
```

```
Number of floating hierarchical blocks: 0
```

```
*****
```

```
*****Verify net VSS connectivity*****
```

```
Number of floating wires: 36
```

```
Number of floating vias: 0
```

```
Number of floating std cells: 43061
```

```
Number of floating hard macros: 0
```

```
Number of floating I/O pads: 0
```

```
Number of floating terminals: 0
```

```
Number of floating hierarchical blocks: 0
```

```
*****
```

```
Overall runtime: 0 seconds.
```

```
{PATH_11_493 PATH_11_454 PATH_11_457 PATH_11_460 PATH_11_463 PATH_11_466 PATH_11_469 PATH_11_472 PATH_11_475 PATH_11_478 PATH_11_481 PATH_11_484 PATH_11_487 PATH_11_490 PATH_11_496 PATH_11_499 PATH_11_502 PATH_11_505 PATH_11_508 PATH_11_511 PATH_11_514 PATH_11_517 PATH_11_520 PATH_11_523 PATH_11_526 PATH_11_611 PATH_19_1959 PATH_19_1967 PATH_19_1966 PATH_19_1965 PATH_19_1964 PATH_19_1963 PATH_19_1962 PATH_19_1961 PATH_19_1960 PATH_19_1979 PATH_19_1958 PATH_19_1957 PATH_19_1956 PATH_19_1955 PATH_19_1954 PATH_19_1953 PATH_19_1952 PATH_19_1951 PATH_19_1988 PATH_19_1996 PATH_19_1995 PATH_19_1994 PATH_19_1993 PATH_19_1992 PATH_19_1991 PATH_19_1990 PATH_19_1989 PATH_19_1968 PATH_19_1987 PATH_19_1986 PATH_19_1985 PATH_19_1984 PATH_19_1983 PATH_19_1982 PATH_19_1981 PATH_19_1980}
```